Run report: tiantheory-run-20261024T2225Z

Descriptive record of one recorded run of examples/genus4_n4_census.rs, part TIANTHEORY under `research/genus4_n4_census_20261024/PROTOCOL.md` (with its dated amendments). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
run: tiantheory 20261024T2225Z
git HEAD: 7a579ce24983ef78f6af4ecf3dc9a8c6f10e8a91
tool: examples/genus4_n4_census.rs sha256 cc0bcbe7b428feff4e2acbea3c6887d6abb5031b01f3a49589bd3597b3773dc7
protocol: research/genus4_n4_census_20261024/PROTOCOL.md sha256 457dc80ceded9acbbb0fb39ca1c61e8c3e2bfa773f863100584969c76131187d
command: /tmp/claude-0/g4/g4 research/genus4_n4_census_20261024/results/tiantheory-run-20261024T2225Z TIANTHEORY 3,5,7
rustc: rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v153 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz
threads: 1
wall seconds: 84
```

tool_log.txt

```
theory check l=3 done
theory check l=5 done
theory check l=7 done
Theorem 6 cells compared: 44; disagreements: 0
```

non_regular_cells.txt

```
l=3 qbar=1 tr=1 scalar: n_ker=12 n_tian=8
l=3 qbar=1 tr=1 jordan: n_ker=3 n_tian=2
l=3 qbar=1 tr=2 scalar: n_ker=64 n_tian=60
l=3 qbar=1 tr=2 jordan: n_ker=1 n_tian=0
l=3 qbar=2 tr=1 scalar: n_ker=16 n_tian=12
l=3 qbar=2 tr=1 jordan: n_ker=1 n_tian=0
l=3 qbar=2 tr=2 scalar: n_ker=60 n_tian=56
l=3 qbar=2 tr=2 jordan: n_ker=3 n_tian=2
l=5 qbar=1 tr=2 scalar: n_ker=288 n_tian=282
l=5 qbar=1 tr=2 jordan: n_ker=3 n_tian=2
l=5 qbar=1 tr=3 scalar: n_ker=28 n_tian=22
l=5 qbar=1 tr=3 jordan: n_ker=3 n_tian=2
l=5 qbar=2 tr=2 scalar: n_ker=64 n_tian=58
l=5 qbar=2 tr=2 jordan: n_ker=9 n_tian=8
l=5 qbar=2 tr=3 scalar: n_ker=156 n_tian=150
l=5 qbar=2 tr=3 jordan: n_ker=1 n_tian=0
l=5 qbar=3 tr=2 scalar: n_ker=64 n_tian=58
l=5 qbar=3 tr=2 jordan: n_ker=9 n_tian=8
l=5 qbar=3 tr=3 scalar: n_ker=156 n_tian=150
l=5 qbar=3 tr=3 jordan: n_ker=1 n_tian=0
l=5 qbar=4 tr=2 scalar: n_ker=288 n_tian=282
l=5 qbar=4 tr=2 jordan: n_ker=3 n_tian=2
l=5 qbar=4 tr=3 scalar: n_ker=28 n_tian=22
l=5 qbar=4 tr=3 jordan: n_ker=3 n_tian=2
l=7 qbar=1 tr=2 scalar: n_ker=512 n_tian=504
l=7 qbar=1 tr=2 jordan: n_ker=1 n_tian=0
l=7 qbar=1 tr=5 scalar: n_ker=64 n_tian=56
l=7 qbar=1 tr=5 jordan: n_ker=1 n_tian=0
l=7 qbar=2 tr=1 scalar: n_ker=512 n_tian=504
l=7 qbar=2 tr=1 jordan: n_ker=1 n_tian=0
l=7 qbar=2 tr=6 scalar: n_ker=64 n_tian=56
l=7 qbar=2 tr=6 jordan: n_ker=1 n_tian=0
l=7 qbar=3 tr=3 scalar: n_ker=52 n_tian=44
l=7 qbar=3 tr=3 jordan: n_ker=3 n_tian=2
l=7 qbar=3 tr=4 scalar: n_ker=500 n_tian=492
l=7 qbar=3 tr=4 jordan: n_ker=3 n_tian=2
l=7 qbar=4 tr=3 scalar: n_ker=64 n_tian=56
l=7 qbar=4 tr=3 jordan: n_ker=1 n_tian=0
```

```
l=7 qbar=4 tr=4 scalar: n_ker=512 n_tian=504
l=7 qbar=4 tr=4 jordan: n_ker=1 n_tian=0
l=7 qbar=5 tr=1 scalar: n_ker=500 n_tian=492
l=7 qbar=5 tr=1 jordan: n_ker=3 n_tian=2
l=7 qbar=5 tr=6 scalar: n_ker=52 n_tian=44
l=7 qbar=5 tr=6 jordan: n_ker=3 n_tian=2
l=7 qbar=6 tr=2 scalar: n_ker=500 n_tian=492
l=7 qbar=6 tr=2 jordan: n_ker=3 n_tian=2
l=7 qbar=6 tr=5 scalar: n_ker=52 n_tian=44
l=7 qbar=6 tr=5 jordan: n_ker=3 n_tian=2
```

summary.txt

```
Theorem 6 cells compared: 44; disagreements: 0
```

tian_theory.csv (first 12 of 92 rows)

ell	qbar	tr	det	class	theory	linear_algebra	n_ker	agree
3	1	0	1	nonsplit	2	2	2	true
3	1	1	1	scalar	-	8	12	true
3	1	1	1	jordan	-	2	3	true
3	1	2	1	scalar	-	60	64	true
3	1	2	1	jordan	-	0	1	true
3	2	0	1	nonsplit	2	2	2	true
3	2	1	1	scalar	-	12	16	true
3	2	1	1	jordan	-	0	1	true
3	2	2	1	scalar	-	56	60	true
3	2	2	1	jordan	-	2	3	true
5	1	0	1	split	0	0	2	true
5	1	1	1	nonsplit	4	4	4	true

Graph

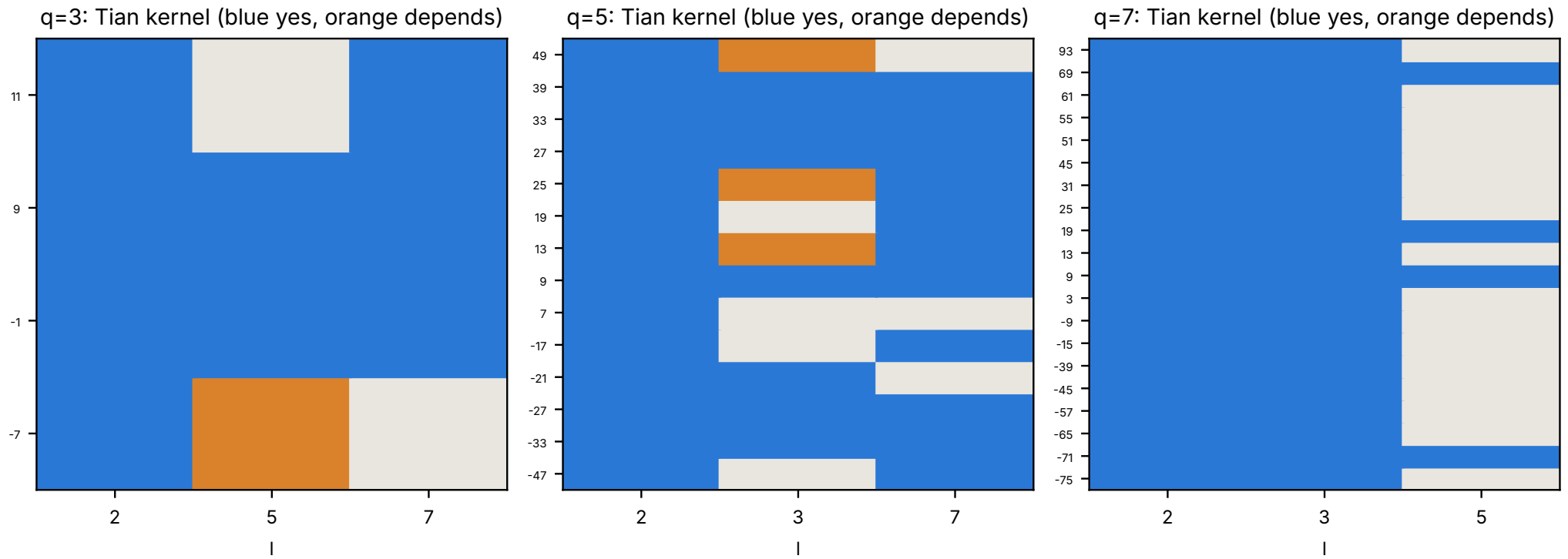


Figure 1: Tian kernel existence per prime-order trace and l (n = 4).